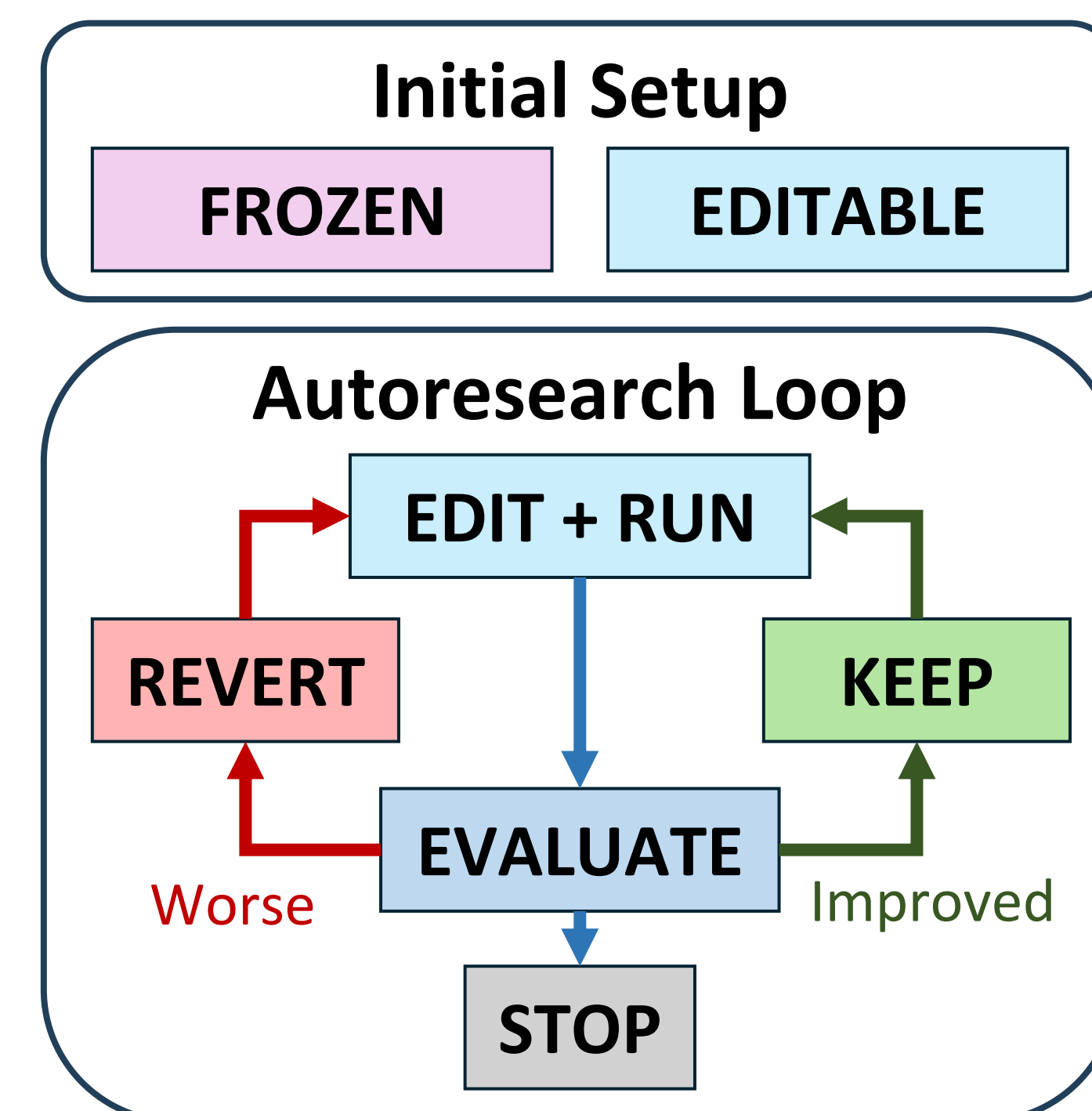


Optimizing dimensionality reduction in gene expression using autoresearch

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Background

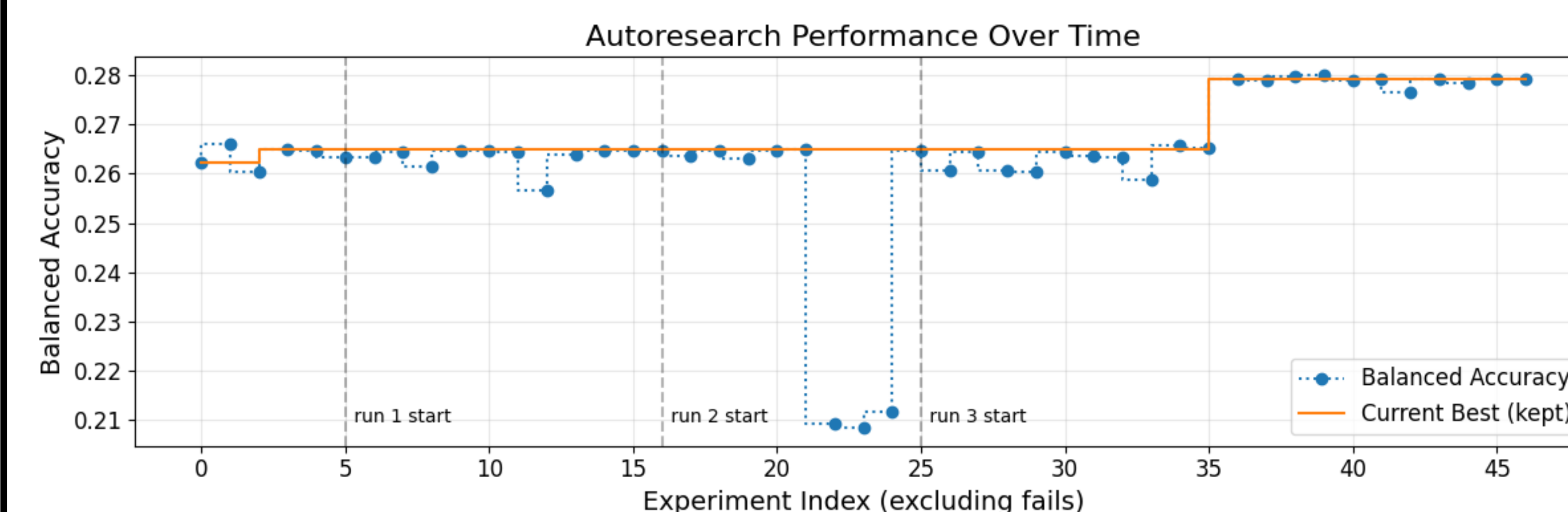
- *Autoresearch* [1]: Use an AI agent within a controlled framework to conduct autonomous experiments, optimizing results according to a fixed evaluation metric
- Efficient alternative to manual model specification and hyperparameter tuning that harnesses strengths of agentic AI
- **How can autoresearch aid in dimension reduction for isolating biological signal from noise?**



- High-dimensional gene expression data can often be challenging to interpret due to the “curse of dimensionality”
- Dimension reduction can separate biologically significant signals from sources of insignificant noise
- Low-dimensional embedding structure could reveal gene relationships that underlie higher level behaviors (phenotype)

Results

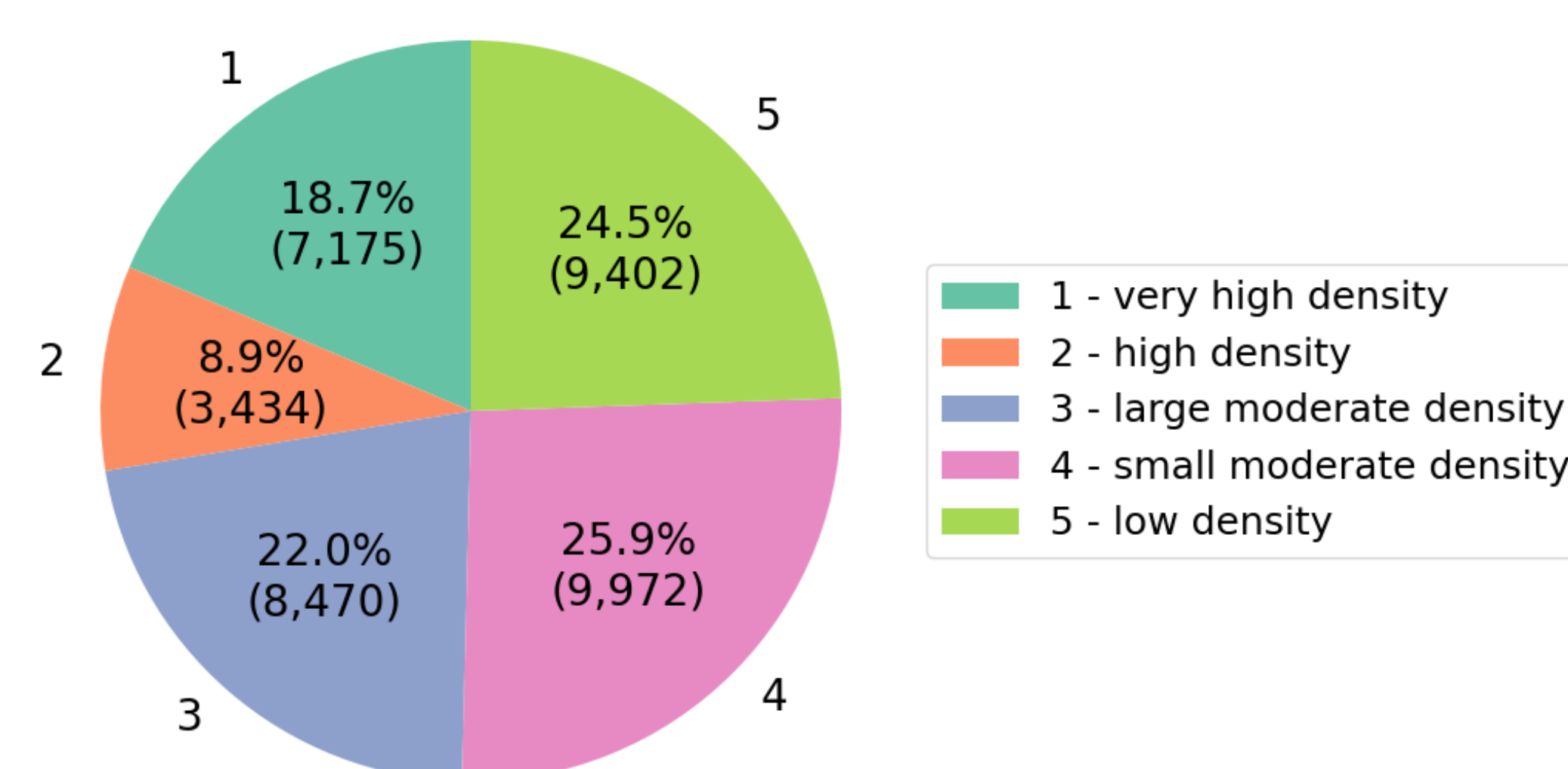
- **4** autoresearch runs; **58** autonomous experiments (**2** baseline, **3** kept, **42** discarded, **11** failed); **10+** dimension reduction approaches tested
- **1.7%** improvement in validation accuracy over the baseline



Methods

- Data from Allen Brain Atlas: Microarray data of gene expression from the human brain (~60,000 gene probes across 946 brain regions) [2]
- Goal to locate the chromosome on which each gene probe lies (multiclass classification)
- Relabel target variable into chromosome groupings to reduce the number of unique levels (24 to 5)

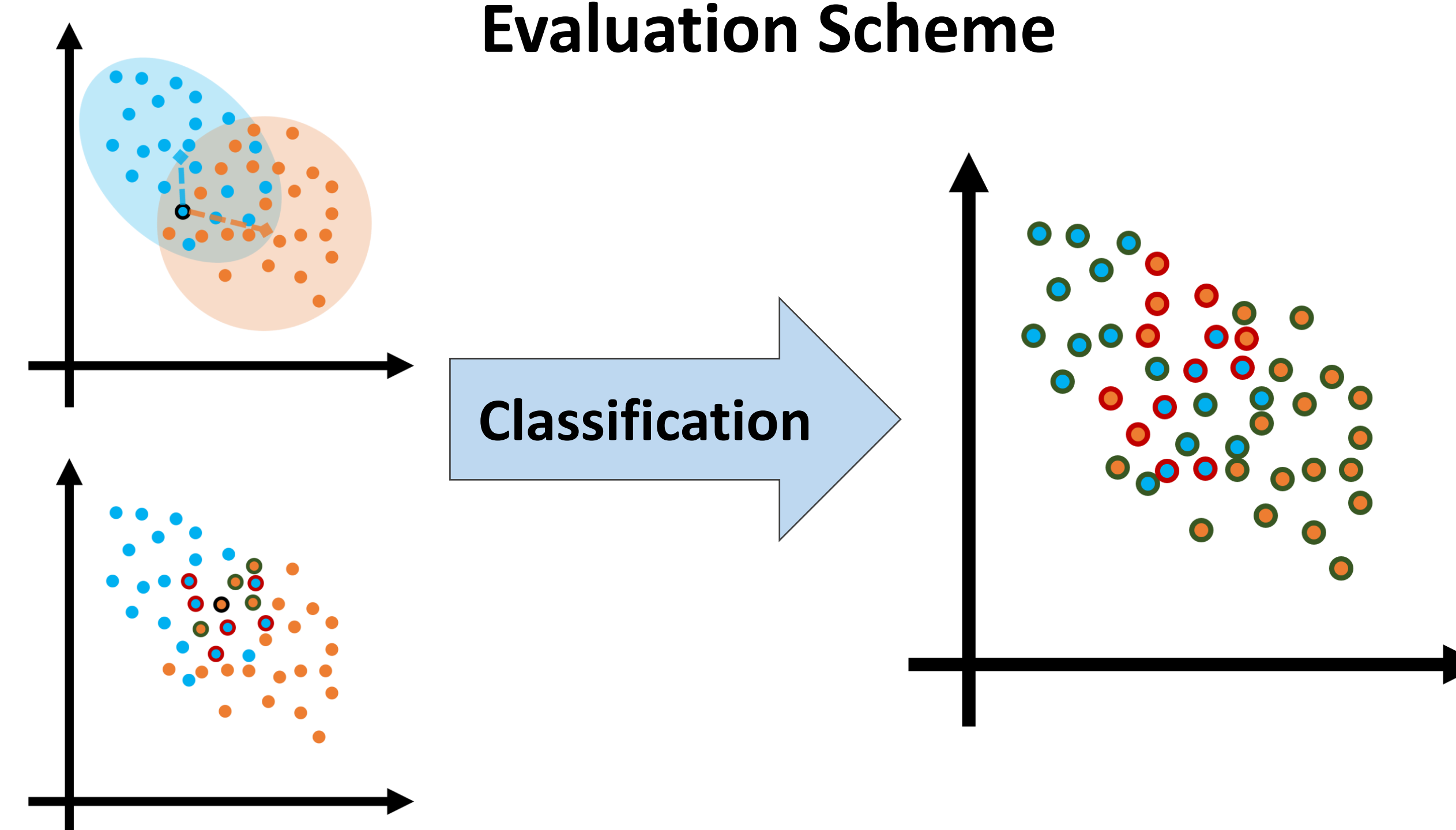
Distribution of Chromosome Groupings



Core Autoresearch Components

- AI Agent: Large Language Model that directly interacts with and modifies the codebase (Codex, Claude Code)
- Agent Skills: Reusable instructions that assist AI agents in performing specialized tasks (scientific notetaking)
- FROZEN modules: *prepare.py*, *program.md*
- EDITABLE modules: *model.py*, experiment logs

Evaluation Scheme



- Silhouette score: Evaluate separability of class labels in low-dimensional space (-1 to 1); $s = \frac{b_i - a_i}{\max(a_i, b_i)}$
- Precision@k ($k = 10$): Evaluate local information retrieval from low-dimensional embeddings (0 to 1)
- Balanced accuracy: Evaluate classification performance following dimension reduction (0 to 1)
- *Keep* experiments with improvements over a threshold to avoid overfitting; *Discard* if performance worsens

Takeaways

- This project successfully implemented autoresearch to exhaustively explore a large search space during the model selection process
- The classification problem is inherently challenging, making it difficult to attain a strong objective classification performance
- Future work could improve performance by incorporating specific domain knowledge within the agent decision-making framework
- Other applications of autoresearch could better highlight its capability to produce meaningful progress and performance improvements

Run #	Exp #	Description	Sil Score	Prec@10	Acc	Runtime	Status
0	0	Baseline: PCA(250) + LR	-0.030370	0.235002	0.262215	5.51s	baseline
0	4	FastICA	-0.008012	0.241386	0.265090	205.55s	keep
1	2	QuantileTransformer + FastICA	-0.006324	0.237394	0.263286	199.98s	discard
1	9	Autoencoder	-0.023219	0.238669	0.256687	424.04s	discard
2	9	UMAP	-0.046513	0.227214	0.209294	135.85s	discard
3	14	FastICA + LR(class_weight=balanced)	-0.008012	0.241386	0.279145	73.94s	keep

References

- [1] A. Karpathy, “autoresearch,” GitHub repository. Accessed: May 18, 2026. [Online]. Available: <https://github.com/karpathy/autoresearch>
- [2] M. J. Hawrylycz et al., “An anatomically comprehensive atlas of the adult human brain transcriptome,” *Nature*, vol. 489, no. 7416, pp. 391–399, Sep. 2012, doi: 10.1038/nature11405.

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